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Deep Learning (285652)

From Scratch to Pretrained: When Does Pretraining Pay Off?

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1 Abstract

This report examines when pretraining pays off for IMDB sentiment classification under different labeled-data budgets. We compare five approaches on the IMDB Large Movie Review Dataset [6]: an LSTM trained from scratch [1], an LSTM with frozen GloVe embeddings [2], a BiLSTM with GloVe and attention pooling, fine-tuned DistilBERT [4, 3], and GPT-2 prompting with zero-shot and few-shot prompts [5]. Training sizes are $N \in \{50, 200, 1000, 5000, 25000\}$ with three balanced random subsets per size. We decompose the improvement from a from-scratch LSTM to DistilBERT into three effects: representation quality, architecture/pooling, and contextual pretraining. The main crossover occurs between $N = 50$ and $N = 200$: at $N = 50$, BiLSTM+GloVe+Attn is marginally the strongest trained model, while from $N = 200$ onward DistilBERT is clearly strongest. Due to runtime constraints, GPT-2 prompting is reported as a separate approximate zero-training baseline on the first 200 IMDB test examples, with unparsed outputs excluded from the accuracy and macro-F1 computation; therefore, prompting results should be interpreted only as an approximate auxiliary baseline. Practically, BiLSTM+GloVe+Attn is attractive at 50 labels or under strict latency constraints, while DistilBERT is preferable from 200 labels onward if compute is available.

2 Introduction

Labeled data is often the limiting factor in applied NLP. For a new domain, a practitioner may have only a few dozen or a few hundred annotated examples, making it unclear whether to train a small model from scratch, use static pretrained word vectors, fine-tune a contextual model, or rely on prompting. This report studies that decision through a controlled comparison of increasingly pretrained methods.

The research question is: how do a from-scratch LSTM, an LSTM with GloVe embeddings, a BiLSTM with GloVe and attention pooling, fine-tuned DistilBERT, and prompted GPT-2 compare on sentiment classification, and how much does each improvement contribute? We structure the analysis around three gaps: the representation effect from LSTM to LSTM+GloVe, the architecture/pooling effect from LSTM+GloVe to BiLSTM+GloVe+Attn, and the contextual pretraining premium from BiLSTM+GloVe+Attn to DistilBERT.

We expect static GloVe embeddings to help when labels are scarce, but less than contextual pretraining. We also expect bidirectionality and attention pooling to improve over last-state LSTM aggregation for long reviews. DistilBERT should become strongest once enough labeled data is available for stable fine-tuning, while GPT-2 prompting is expected to be useful mainly as a zero-training baseline rather than as a replacement for supervised fine-tuning.

3 Methods

3.1 Dataset and evaluation protocol

All experiments use the IMDB Large Movie Review Dataset [6]. Training sizes are 50, 200, 1000, 5000, and 25000. For each reduced size, balanced subsets with equal positive and negative examples are sampled using three seeds. Accuracy and macro-F1 are reported as mean $\pm$ standard deviation over these seeds. LSTM-family models and DistilBERT are evaluated on the full IMDB test split. No validation split or early stopping is used; models are trained for fixed epochs. Runtime measurements were collected in the same CUDA-enabled environment for all trained models.

Because GPT-2 prompting was computationally much slower, it was evaluated as an approximate auxiliary zero-training baseline on the first 200 examples of the IMDB test split. Therefore, prompting results are not directly comparable to the trained classifiers evaluated on the full IMDB test set. The prompting results describe GPT-2 in this local setup and should not be generalized to modern instruction-tuned LLMs. This is a limitation and a deviation from the ideal protocol specified in the assignment, where all approaches would be evaluated on the full IMDB test split. We keep the prompting results in the report because they still provide a useful zero-training reference point, but we do not use them for direct statistical comparisons with the trained models.

3.2 Models

The from-scratch LSTM uses randomly initialized 100-dimensional embeddings, hidden size 128, one LSTM layer, dropout 0.5, and a linear classifier [1]. Reviews are tokenized with a vocabulary size of 20000 and padded or truncated to length 200. The model is trained for 10 epochs with Adam, learning rate 10^{-3} , batch size 64, cross-entropy loss, and gradient clipping.

LSTM+GloVe keeps the same architecture but initializes the embedding layer with frozen 100-dimensional GloVe vectors [2]. BiLSTM+GloVe+Attn uses frozen GloVe embeddings, a bidirectional LSTM with hidden size 128 per direction, and learned attention pooling over hidden states. DistilBERT fine-tunes `distilbert-base-uncased` for binary classification with maximum sequence length 256, 3 epochs, learning rate 2×10^{-5} , training batch size 16, evaluation batch size 32, and weight decay 0.01 [4, 3]. GPT-2 prompting uses zero-shot, 2-shot, 4-shot, and 8-shot prompts. Few-shot examples are demonstrations in the prompt, not training examples for weight updates. Generated labels are parsed as positive or negative. Outputs that could not be parsed were excluded from the accuracy and macro-F1 computation, and the parse rate is reported separately.

4 Results

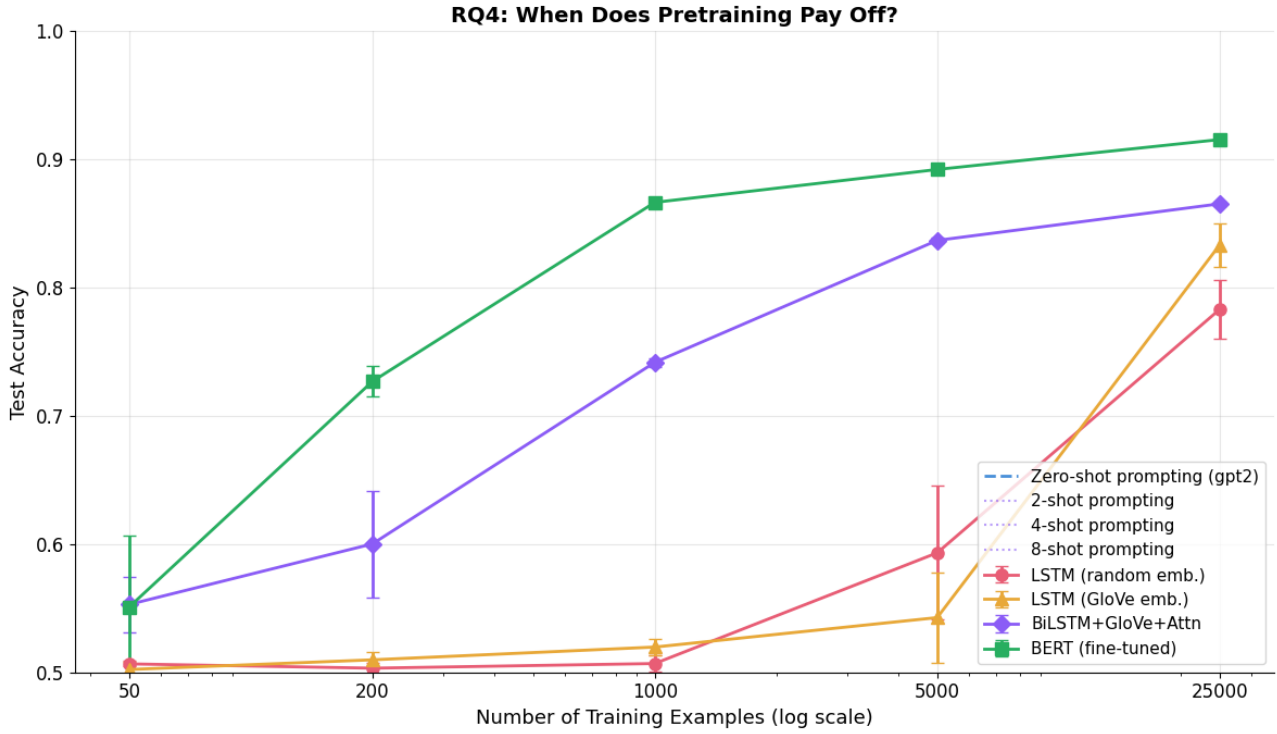


Figure 1: Accuracy across training-set sizes. LSTM, LSTM+GloVe, BiLSTM+GloVe+Attn, and DistilBERT are trained over N and evaluated on the full IMDB test split. GPT-2 prompting does not use N training examples or weight updates; its zero-shot and few-shot results are shown only as approximate auxiliary zero-training baselines based on the first 200 examples of the IMDB test split.

Figure 1 shows the main crossover. At $N = 50$, BiLSTM+GloVe+Attn is marginally the strongest trained model, with 0.5529 accuracy compared with 0.5510 for DistilBERT. From $N = 200$ onward, DistilBERT is the strongest trained model. At $N = 200$, DistilBERT reaches 0.7269 accuracy, exceeding BiLSTM+GloVe+Attn by 0.1267. The advantage remains large at $N = 1000$ (+0.1247) and smaller but consistent at $N = 5000$ and $N = 25000$.

Model	N	Accuracy	Macro-F1	Train time (s)	Inf./ex. (s)
LSTM	50	0.5066 ± 0.0021	0.4577 ± 0.0331	0.1	0.000027
LSTM	200	0.5033 ± 0.0041	0.4608 ± 0.0150	0.1	0.000027
LSTM	1000	0.5069 ± 0.0061	0.4691 ± 0.0051	0.6	0.000026
LSTM	5000	0.5932 ± 0.0520	0.5760 ± 0.0672	2.8	0.000027
LSTM	25000	0.7831 ± 0.0226	0.7829 ± 0.0225	14.8	0.000027
LSTM+GloVe	50	0.5022 ± 0.0059	0.5000 ± 0.0060	0.0	0.000030
LSTM+GloVe	200	0.5098 ± 0.0063	0.4451 ± 0.0521	0.1	0.000032
LSTM+GloVe	1000	0.5197 ± 0.0063	0.4567 ± 0.0064	0.5	0.000029
LSTM+GloVe	5000	0.5427 ± 0.0350	0.4675 ± 0.0436	2.5	0.000028
LSTM+GloVe	25000	0.8333 ± 0.0170	0.8324 ± 0.0182	12.5	0.000028
BiLSTM+GloVe+Attn	50	0.5529 ± 0.0216	0.5220 ± 0.0356	0.1	0.000045
BiLSTM+GloVe+Attn	200	0.6002 ± 0.0415	0.5662 ± 0.0804	0.2	0.000045
BiLSTM+GloVe+Attn	1000	0.7418 ± 0.0035	0.7388 ± 0.0023	1.0	0.000046
BiLSTM+GloVe+Attn	5000	0.8369 ± 0.0023	0.8367 ± 0.0024	5.1	0.000045
BiLSTM+GloVe+Attn	25000	0.8654 ± 0.0016	0.8653 ± 0.0016	25.7	0.000045
DistilBERT	50	0.5510 ± 0.0559	0.4551 ± 0.1241	1.0	0.001211
DistilBERT	200	0.7269 ± 0.0119	0.7250 ± 0.0116	3.2	0.001225
DistilBERT	1000	0.8665 ± 0.0026	0.8665 ± 0.0027	14.5	0.001214
DistilBERT	5000	0.8922 ± 0.0029	0.8922 ± 0.0029	72.0	0.001232
DistilBERT	25000	0.9155 ± 0.0000	0.9155 ± 0.0000	367.5	0.001260

Table 1: Main results for trained models. Accuracy and macro-F1 are reported as mean $\pm$ standard deviation over three seeds. Training time is reported in seconds, and inference time is reported in seconds per example.

Table 1 shows that GloVe alone gives only small and inconsistent gains over the random LSTM. The main low-data improvement comes from BiLSTM+GloVe+Attn, especially at $N = 200$. DistilBERT is already clearly superior at $N=200$ and remains strongest at $N=1000$, $N=5000$, and $N=25000$.

N	Representation	Architecture/Pooling	Contextual Pretraining
	LSTM+GloVe – LSTM	BiLSTM+GloVe+Attn – LSTM+GloVe	DistilBERT – BiLSTM+GloVe+
50	−0.0044	+0.0507	−0.0020
200	+0.0065	+0.0904	+0.1267
1000	+0.0128	+0.2221	+0.1247
5000	−0.0505	+0.2942	+0.0553
25000	+0.0503	+0.0321	+0.0501

Table 2: Accuracy gap decomposition.

The gap table makes the central finding explicit. The representation effect from random embeddings to frozen GloVe is weak and inconsistent: it is slightly negative at $N = 50$ and $N = 5000$, but positive at $N = 200$, $N = 1000$, and $N = 25000$. The architecture/pooling effect is consistently positive and is largest at $N = 5000$. The contextual pretraining premium is not visible at $N = 50$, where DistilBERT is slightly below BiLSTM+GloVe+Attn, but becomes strong from $N = 200$ onward.

Model	Shots	Accuracy	Macro-F1
Prompt-ZS	0	0.3333 ± 0.0000	0.2500 ± 0.0000
Prompt-2shot	2	0.2987 ± 0.0158	0.2299 ± 0.0094
Prompt-4shot	4	0.3398 ± 0.0511	0.2525 ± 0.0290
Prompt-8shot	8	0.4241 ± 0.0390	0.2973 ± 0.0196

Table 3: GPT-2 prompting results on the first 200 examples of the IMDB test split. Unparsed generations are excluded from accuracy and macro-F1 computation. Because prompting was not evaluated on the full 25,000-example test split, these results are approximate auxiliary baselines and are not directly comparable to the trained models in Table 1.

As shown in Table 3, GPT-2 prompting was evaluated only on the first 200 IMDB test examples rather than the full IMDB test set. It is therefore treated as a separate approximate zero-training baseline rather than as a fully comparable main benchmark condition. In this setup, GPT-2 prompting did not compete with supervised fine-tuning. However, this result should not be generalized to modern instruction-tuned LLMs, which may perform substantially better with carefully designed prompts. The prompting results are also affected by parse failures, especially in the zero-shot setting; few-shot prompts improved the reliability of the generated labels in our implementation, but the results remain only an approximate auxiliary baseline.

5 Discussion

The results answer the research question through the three gaps in Table 2. Static representation pretraining alone is not enough: LSTM+GloVe improves over the random LSTM only weakly at $N = 200$ (+0.0065) and $N = 1000$ (+0.0128), is slightly worse at $N = 50$ and $N = 5000$, and gives a larger gain only in the full-data setting (+0.0503). In contrast, the architecture/pooling step is consistently positive. At $N = 200$, BiLSTM+GloVe+Attn improves over LSTM+GloVe by +0.0904 accuracy, and the largest architecture/pooling gain appears at $N = 5000$ (+0.2942). This suggests that bidirectionality and attention pooling address a major weakness of the last-state LSTM on long movie reviews.

The contextual pretraining premium appears from $N = 200$ onward. At $N = 50$, DistilBERT is slightly below BiLSTM+GloVe+Attn in mean accuracy. At $N = 200$, however, DistilBERT is

already clearly ahead by +0.1267 accuracy. The advantage remains large at $N = 1000$ (+0.1247) and smaller but consistent at $N = 5000$ (+0.0553) and $N = 25000$ (+0.0501). On the full dataset, DistilBERT reaches 0.9155 accuracy, compared with 0.8654 for BiLSTM+GloVe+Attn.

The cost-benefit trade-off is substantial. At $N = 25000$, DistilBERT requires 367.5 seconds of training time, compared with 25.7 seconds for BiLSTM+GloVe+Attn. DistilBERT inference is also much slower per example: 0.001260 seconds compared with 0.000045 seconds for BiLSTM+GloVe+Attn in these measurements. Thus, DistilBERT is preferable when peak accuracy matters and compute is available, while BiLSTM+GloVe+Attn remains attractive under strict latency or cost constraints.

5.1 Error analysis

The exported error examples at $N = 200$ suggest that DistilBERT fixes some errors made by the LSTM when the sentiment depends on broader context rather than isolated sentiment words. In several cases, the LSTM appears to overreact to local negative or positive cues, while DistilBERT better captures the overall polarity of the review.

Prompting also correctly classifies some examples that are missed by the trained models, but these cases should be interpreted cautiously because the prompting evaluation was performed only on the first 200 test examples and unparsed generations were excluded.

The remaining shared errors indicate systematic difficulties with sarcasm, mixed sentiment, delayed sentiment reversal, and reviews that describe positive reception before expressing a negative final judgment. These patterns suggest that attention and contextual pretraining help with long-range context, but do not fully solve pragmatic ambiguity or implicit sentiment.

5.2 RQ3 and practical recommendation

The RQ4 results also clarify the earlier RQ3 comparison with TextCNN. In RQ3, the best TextCNN result was obtained at sequence length 512, with 0.8710 accuracy and 0.8710 macro-F1. In the full-data RQ4 setting, BiLSTM+GloVe+Attn reaches 0.8654 accuracy and 0.8653 macro-F1. Thus, the improved recurrent model nearly matches, but does not quite exceed, the earlier TextCNN result. This suggests that the weakness of the plain LSTM in RQ3 was not only a representation problem: frozen GloVe embeddings alone help only modestly, whereas bidirectionality and attention pooling close most of the gap to TextCNN. The remaining small advantage of TextCNN is consistent with the IMDB task, where local sentiment cues and short evaluative phrases are often highly informative.

The practical recommendation is therefore data-dependent. With 0 labels, prompting is possible, but GPT-2 is only a weak local baseline in this setup. With 50 labels, BiLSTM+GloVe+Attn is the strongest trained model and offers very low inference cost. From 200 labels onward, DistilBERT is the best choice if compute is available. Under strict latency or cost constraints,

BiLSTM+GloVe+Attn remains a strong compromise. In the full-data setting, DistilBERT gives the best accuracy and macro-F1.

The main limitations are that GPT-2 is not an instruction-tuned LLM, prompting was evaluated only on the first 200 IMDB test examples, unparsed prompting outputs were excluded from the metric computation, hyperparameters were not extensively optimized, and results may not transfer directly from binary IMDB sentiment classification to shorter texts, other domains, or multi-class tasks.

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